CS 380: ARTIFICIAL INTELLIGENCE

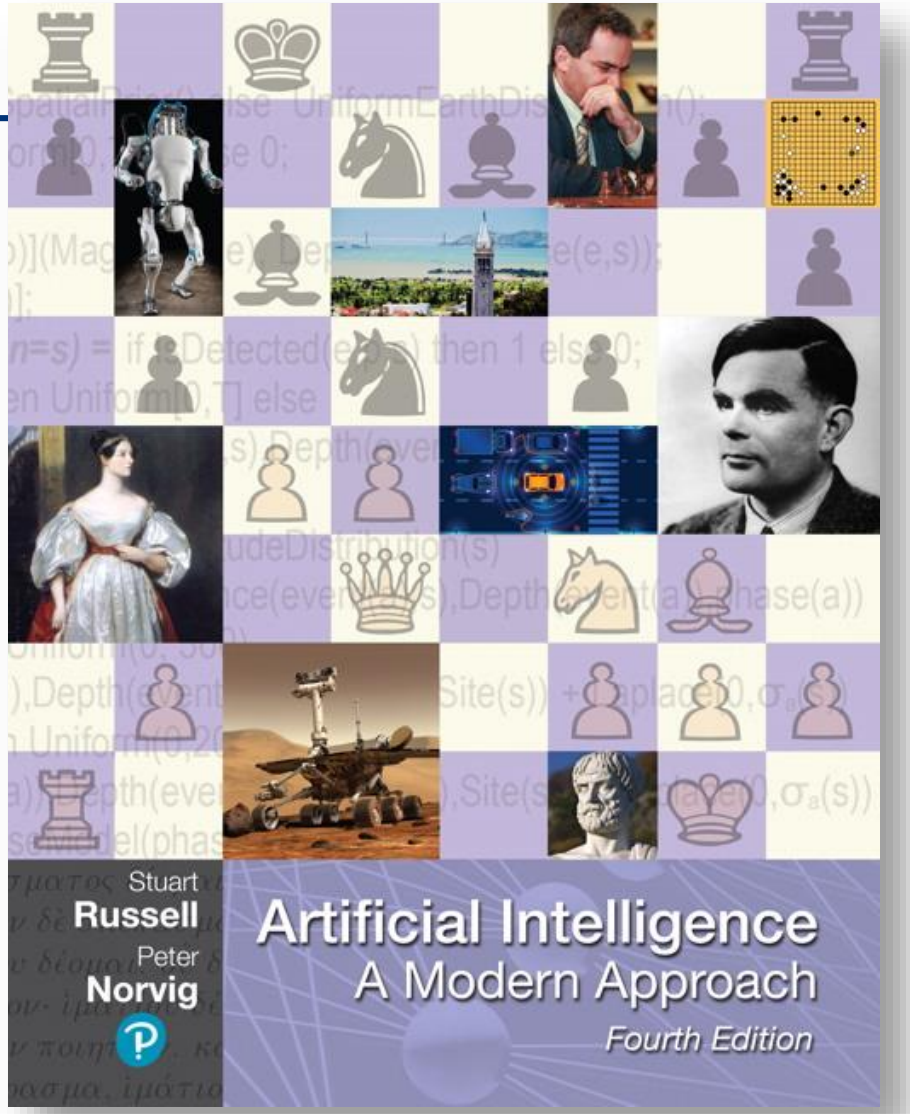
Lecture 1: Introduction

Welcome!

- About me
  - **Jenn Engimann**, Adjunct Professor, Computer Science
  - Email: < jae63@drexel.edu >
  - Office Hours: Thurs 10-11 on Zoom, or by appointment
- Our course TAs
  - **Binh Le** < bql23@drexel.edu >
  - **Don Enwerem** < de435@drexel.edu >
  - **Zaafir ul Hasan** < zuh23@drexel.edu >
  - CLC lab hours posted on Blackboard
- About this course
  - Syllabus, schedule, & resources on Blackboard

Textbook

- Russell & Norvig, 4th edition
  - 3rd is ok too
- A great standard introduction to AI



Classes

- We will have classes twice/week
  - Lecture will be recorded on Echo 360 and posted to Blackboard
- For lectures slides, many thanks to…
  - Dario Salvucci and Santi Ontañón for great material
  - Stuart Russell and Hwee Tou Ng for more great material linked from http://aima.cs.berkeley.edu/
  - The slides here are a conglomeration of all the above plus lots of new material and revisions

Homework Assignments

- 4 programming assignments
  - To be done individually – standard CS rules apply
- For each assignment, you should submit a ZIP file with all requested materials
  - ZIP… not RAR or any other format
  - You may include a README with additional info
  - Failure to follow these guidelines → reduced credit
  - Late submissions lose 33% per day late

Homework Assignments

- Our language of choice: Python 3
  - AI algorithms can be implemented in anything, of course
  - Python is one of the most commonly used languages, and has excellent associated AI/ML libraries
- We will assume that assignments should run on the equivalent of tux.cs.drexel.edu
  - Currently, tux uses Python 3.10.12
- Comment your code when/where needed
  - Not everywhere, just where it's needed
  - We want to understand what you wrote and why, to properly give you credit for what you did

Course Structure: Evaluation

- Homework assignments: 50%
- Midterm exam: 25%
- Final exam: 25%

Blackboard

- All materials will be posted on Blackboard
  - Including schedule, lecture slides, assignments, etc.
- Announcements will generally also be sent out via email

What is Artificial Intelligence?

What is Artificial Intelligence?

Me

Describe AI in Shakespearean English



ChatGPT

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Describe AI in Shakespearean English

ChatGPT

O wondrous craft, born of man's hand and mind,
A creature not of flesh, yet swift and kind,
With circuits bright and logic keen it flies,
A mirror of our thoughts 'neath starry skies.

It learns, it speaks, with wisdom vast and true,
Yet lacks the heart, the soul that we pursue.
No joy, no grief, within its core may dwell,
It thinks, it acts, but cannot love or tell.

In our design, 'tis both a boon and bane,
A tool to ease our work, or bring us pain.
So wield it wise, for power in its might
Can turn to dark or lead to paths of light.

What is Artificial Intelligence?

- Origins (John McCarthy 1956):

We propose that a 2-month, 10-man study of artificial intelligence be carried out during the summer of 1956 at Dartmouth College in Hanover, New Hampshire. The study is to proceed on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it. An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves. We think that a significant advance can be made in one or more of these problems if a carefully selected group of scientists work on it together for a summer.

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Related Fields

▪ Philosophy	Logic, methods of reasoning, mind as physical system foundations of learning, language, rationality
▪ Mathematics & CS Theory	Formal representation and proof algorithms, computation, (un)decidability, (in)tractability, probability
▪ Economics	utility, decision theory
▪ Neuroscience	physical substrate for mental activity
▪ Psychology	phenomena of perception and motor control, experimental techniques, cognitive models
▪ Computer engineering	building fast computers, software vs. hardware
▪ Control theory	design systems that maximize an objective function over time
▪ Linguistics	knowledge representation, grammar

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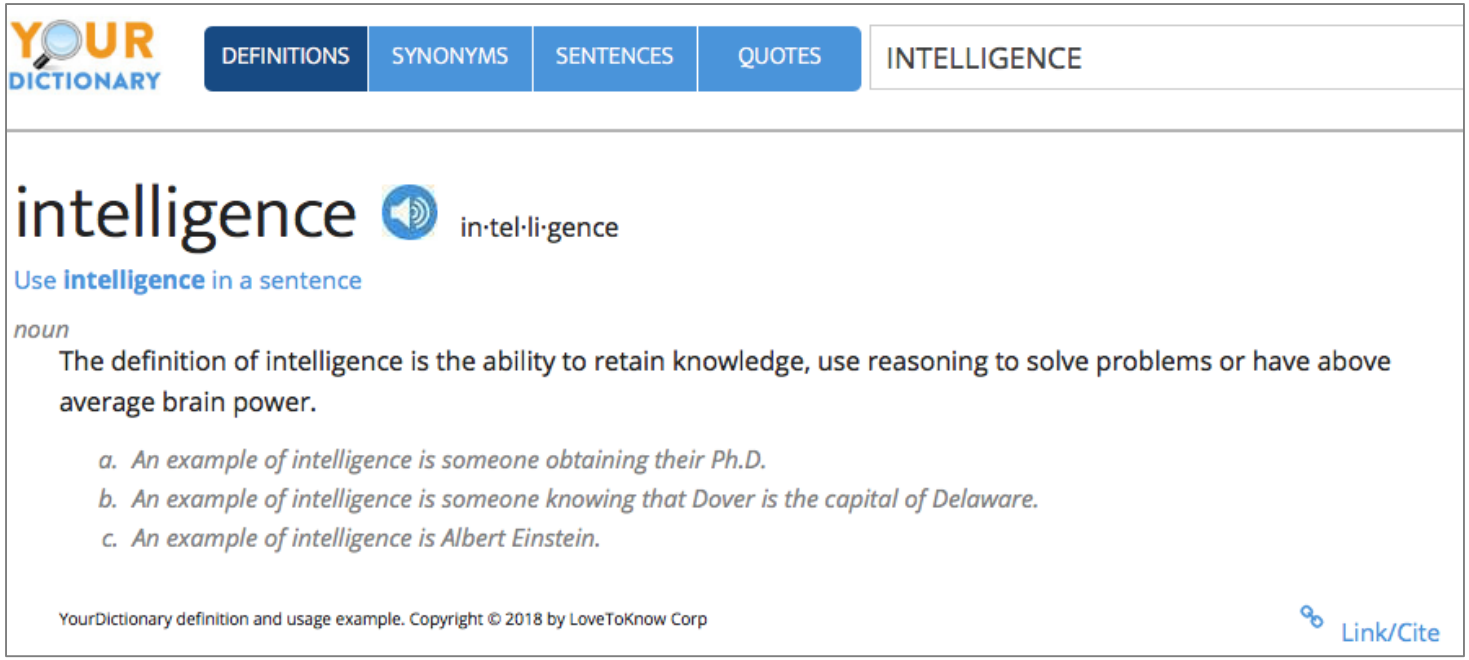
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What is Artificial Intelligence?

- Basically:
  - Replicating (human) intelligence in a machine.
- So, what is “intelligence”?
  - It's very hard to define “intelligence” rigorously
  - Any definition would be arbitrary:
    - We could list features of intelligence, such as: learning, language, knowledge, reasoning, understanding, etc. But any such list leaves things out.
  - Intelligence is often a term that we use to describe certain behaviors that we observe in humans and not in other entities

What is Artificial Intelligence?

- So, if we cannot define intelligence, then… What is AI? And how would we recognize it when we achieve it?

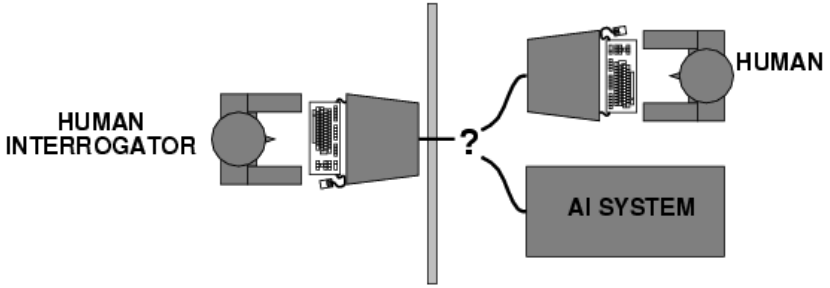
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Thinking humanly	Thinking rationally
Acting humanly	Acting rationally

Acting Humanly: The Turing Test

- Turing (1950) "Computing machinery and intelligence":
- "Can machines think?" → "Can machines behave intelligently?"
- Operational test for intelligent behavior: the Imitation Game



- Turing predicted that by 2000, a machine might have a 30% chance of fooling a lay person for 5 minutes
- Anticipated many of the major arguments for & against AI that would appear over next 50 years
- Suggested major components of AI: knowledge, reasoning, language understanding, learning

Acting Humanly: The Turing Test

- Turing (1950) example:

Q: Please write me a sonnet on the subject of the Forth Bridge.

A : Count me out on this one. I never could write poetry.

Q: Add 34,957 to 70,764.

A: (Pause about 30 seconds and then give as answer) 105,621.

Q: Do you play chess?

A: Yes.

Q: I have K at [e1], and no other pieces. You have only K at [e3] and R at [h8]. It is your move. What do you play?

A: (After a pause of 15 seconds) R-[h1] mate.

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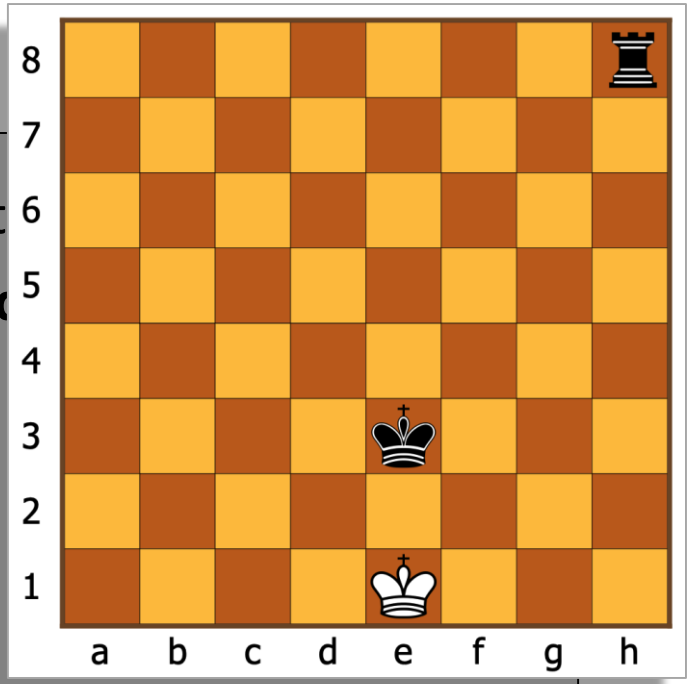
Correct answer: 105,721 !!

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Thinking Humanly: Cognitive Modeling

- 1960s "cognitive revolution": information-processing psychology
- Requires scientific theories of internal activities of the brain
- Validation of these theories would require:
  - Predicting and testing behavior of human subjects (top-down)
  - Direct identification from neurological data (bottom-up)
- Both approaches (roughly, Cognitive Science and Cognitive Neuroscience) are now distinct from AI
  - … but will they merge again??

Thinking Rationally: Laws of Thought

- Aristotle: What are correct arguments/thought processes?
- Several Greek schools developed various forms of logic: notation and rules of derivation for thoughts; may or may not have proceeded to the idea of mechanization
- Direct line through mathematics and philosophy to modern AI
- Problems:
  - Not all intelligent behavior is mediated by logical deliberation
  - What is the purpose of thinking?
    What thoughts should I have?

Acting Rationally: A Rational Agent

- Rational behavior: doing the “right thing”
- The “right thing” = that which is expected to maximize goal achievement, given the available information
- Doesn't necessarily involve thinking – e.g., a blinking reflex – but thinking should be in the service of rational action

Acting Rationally: A Rational Agent

- This course focuses on designing **rational agents**
- An agent is an entity that perceives and acts
- Abstractly, an agent is a function from percept histories to actions:
  - f: P* → A
- For any given class of environments and tasks, we seek the agent (or class of agents) with the best performance
- Caveat: computational limitations make perfect rationality unachievable
  - → design best program for given machine resources

What is Artificial Intelligence?

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Thinking humanly	Thinking rationally
Acting humanly	**Acting rationally**

What is Artificial Intelligence?

- In reality, AI encompasses many computational techniques to replicate any aspect of intelligence:
  - Automated reasoning
  - Knowledge representation
  - Learning
  - Natural language generation & understanding
  - Problem solving
  - Planning, prediction
  - Vision, perception
  - Action
- AI draws on many, many fields: computer science, mathematics, philosophy, economics, psychology, neuroscience, linguistics, control theory, and more

Brief History of AI

• 1943	McCulloch & Pitts: Boolean circuit model of brain
• 1950	Turing's "Computing Machinery and Intelligence"
• 1956	Dartmouth meeting: "Artificial Intelligence" adopted
• 1950s-60s	Early AI programs, including Samuel's checkers program, Newell & Simon's Logic Theorist, Gelernter's Geometry Engine
• 1966-73	AI discovers computational complexity Neural network research almost disappears
• 1969-79	Early development of knowledge-based systems
• 1980--	AI becomes an industry
• 1986--	Neural networks return to popularity
• 1990s	Probability, Agents (neural networks are back)
• 1995--	The emergence of intelligent agents
• 2000s	Human-level AI back on the agenda

What can AI do nowadays?

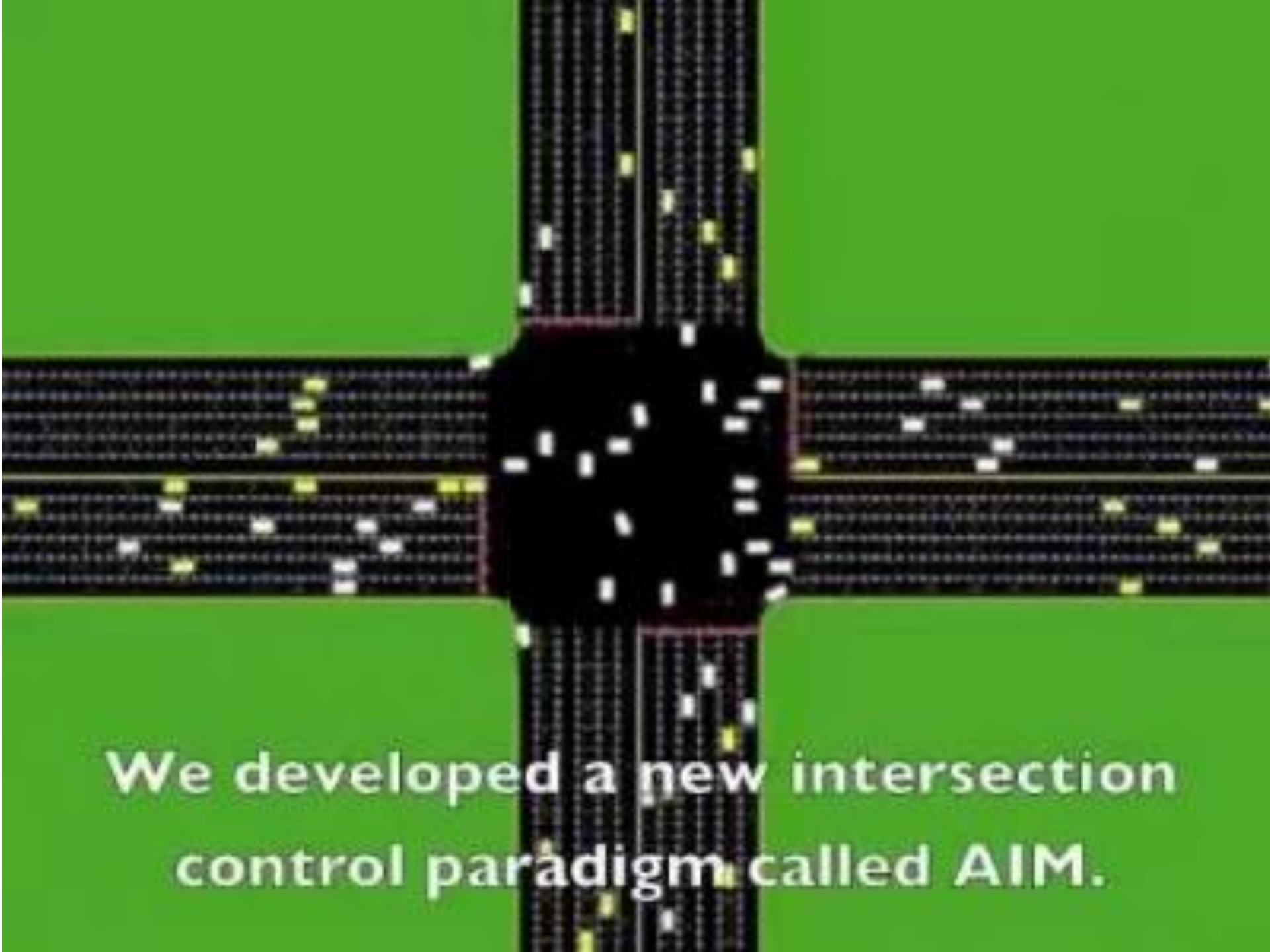
What can AI do nowadays?

- Play ping-pong?
  - https://www.youtube.com/embed/IXyKLDNzGGI



What can AI do nowadays?

- Play ping-pong? YES
- Autonomously drive a car?
  - You all know about autonomous driving
  - Where is the future headed?
    - https://www.youtube.com/embed/4pbAI40dK0A
    - https://www.youtube.com/embed/4SmJP8TdWTU


We developed a new intersection
control paradigm called AIM.



What can AI do nowadays?

- Play ping-pong? YES
- Autonomously drive a car? YES
- Play Soccer?
  - https://www.youtube.com/embed/aLy5pUsmpKE
  - https://www.youtube.com/embed/CJlTj4vmDZE


Carnegie Mellon


1

What can AI do nowadays?

- Play ping-pong? YES
- Autonomously drive a car? YES
- Play Soccer? KIND OF
- Play Chess/Go?
  - https://www.youtube.com/embed/_HCWBS6k8j0

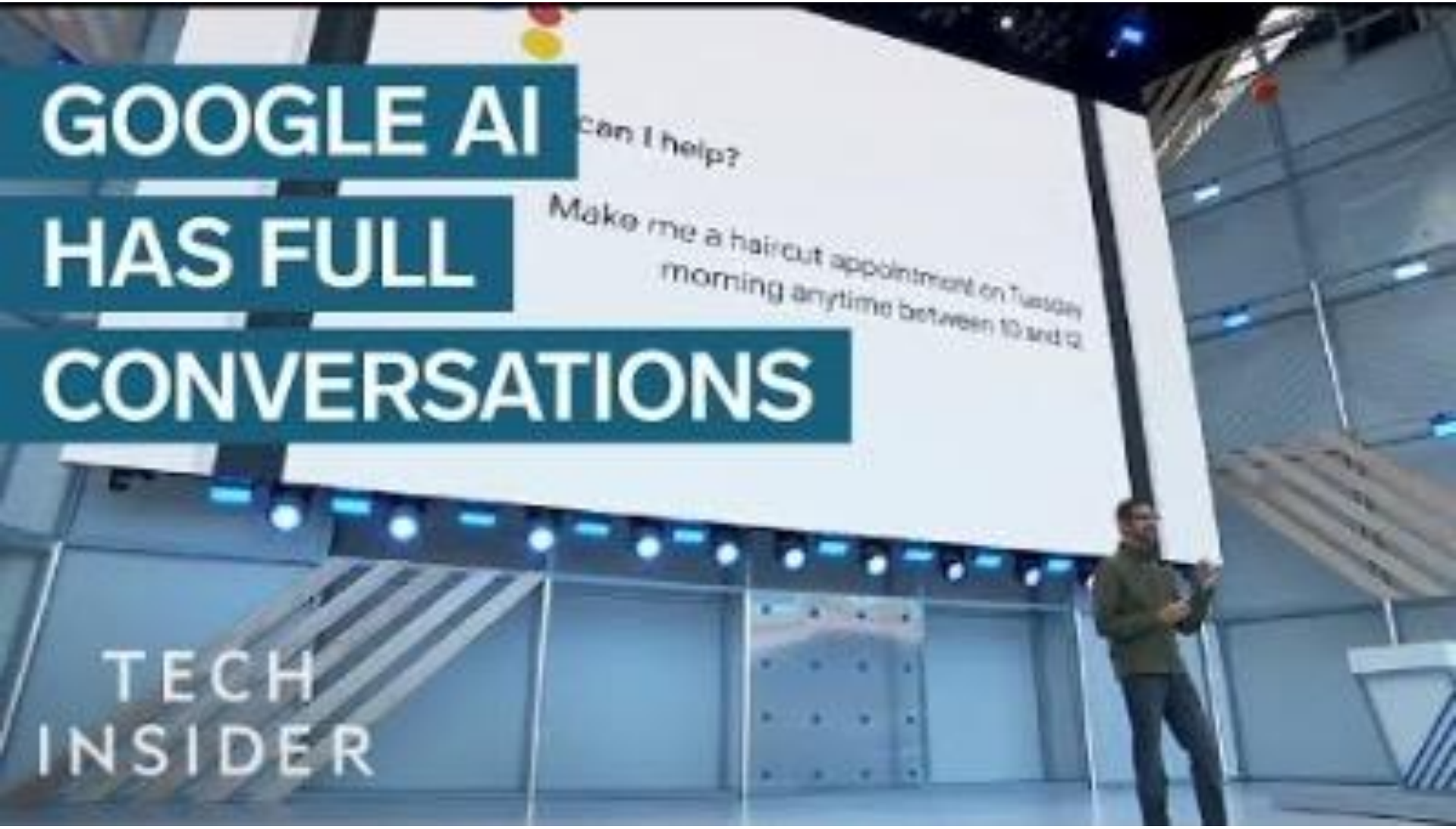

The Future of Go
KE JIE
02:54:50
ALPHAGO
02:57:25

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  - http://translate.google.com

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- Have a conversation?
  - https://www.youtube.com/embed/lXUQ-DdSDoE


GOOGLE AI
HAS FULL
CONVERSATIONS
can I help?
Make me a haircut appointment on Tuesday
morning anytime between 10 and 12
TECH
INSIDER

What can AI do nowadays?

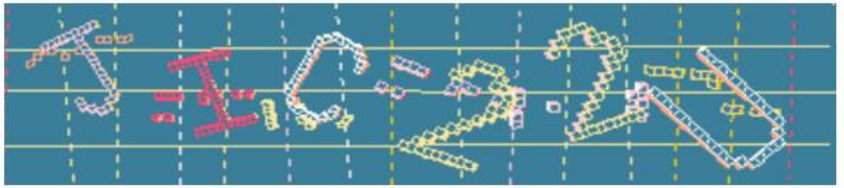
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- Have a conversation? KIND OF
- Understand an image?

What can AI do nowadays?













- Understand an image?

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- Have a conversation? KIND OF
- Understand an image? KIND OF
- Play Jeopardy better than humans?
  - https://www.youtube.com/embed/P18EdAKuC1U


THINK
$24,000
$77,147
$21,600

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- Autonomously drive a car? YES
- Play Soccer? KIND OF
- Play Chess/Go? YES
- Accurately translate a text? KIND OF
- Have a conversation? KIND OF
- Understand an image? KIND OF
- Play Jeopardy better than humans? YES
- Create art?
  - https://www.youtube.com/embed/DJToDLBPovg

AN AVOCADO ARMCHAIR

AI-GENERATED IMAGES



OpenAI

SOTA TEXT-TO-IMAGE

Next Time…

- Rational Agents